

William Cusato

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EDUCATION

Stanford University | Palo Alto, CA

Expected: March 2028

Master of Science in Materials Science and Engineering

Tufts University | Medford, MA

May 2026

Bachelor of Science in Mechanical Engineering with Minor in Computer Science

GPA: 3.97, Dean's List: Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026

Undergraduate Thesis: Investigating the Effect of Deformation and Stirring on Pure Gold Surface Tension

Measurements Using a Faraday Forcing Technique

Departmental Awards: Best Senior Honors Thesis, Lloyd MacGregor Trefethen Fellowship

WORK EXPERIENCE

NASA | Huntsville, AL

June 2026 - August 2026

Materials Science Intern – Electrostatic Levitator Laboratory (ESL) – OSTEM

- Derived properties of ternary and quaternary components of Inconel for use in thermodynamic modelling
- Installed instrumentation of electromagnetic levitator to integrate and communicate directly with LabView
- Utilized SEMs equipped with EDS to take high magnification pictures and determine composition of test samples
- Instructed interns unfamiliar with lab procedures how to run experiments and prepare samples for ESL

Seurat Technologies | Wilmington, MA

June 2025 - August 2025

Parts Production Intern

- Participated in internal and certification audits to gain certification with standards necessary for company growth
- Created and tested Vickers hardness of aluminum test samples for a variety of heat treatments
- Collaborated cross-functionally to write SOPs for complex Powder Bed Fusion 3D Metal Printing machines

Type One Energy Group | Woburn, MA

May 2024 - August 2024

Cryogenic Engineering Intern

- Tested and analyzed superconducting tapes and cables to measure the degradation of critical current.
- Utilized Ansys to simulate thermal strains and stresses on components exposed to cryogenic temperatures.
- Devised cryogenic helium-cooling arrangements for thermal shields that effectively maintain low temperatures.
- Designed a liquid helium distribution system to cool a superconducting magnet to 20 K with team members

Tufts University Department of Engineering | Medford, MA

September 2024 – May 2025

Course Assistant, ME10 - Materials and Manufacturing, ME21 - Dynamics

- Hosted up to 10 hours a week of office hours for students focusing on conceptual problems and hands-on projects
- Attended and participated in regular meetings with fellow course assistants and Tufts University professors

SELECTED PROJECTS

Manufacturing Plan for Stage 2 Turbine Disk of GE CT7 Engine

- Created a manufacturing plan, including non-destructive inspection of turbine disk from original part drawings
- Calculated cost to manufacture turbine disk including tooling, machining, labor, inspection and outsourcing

Investigating Thermal and Processing Effects of HDPE

- Conducted ASTM D638 tensile testing on an Instron 5800R with environmental chamber
- Compared extruded and injection molded HDPE specimens, relating microstructural difference to performance
- Diagnosed a moisture-induced voiding defect in molded samples, tracing root cause to improper pellet drying

Production of Model Prosthetic Running Blade

- Designed and ran multiple Finite Element Analysis simulations to test the impact of a human running stride
- Utilized an Instron Compressive and Tensile Machine to test the validity of FEA simulations
- Examined and discussed failure methods for both composite (Markforged Onyx) and polymer (PLA) filaments

ACTIVITIES

Cross Country / Track and Field | **Tufts University** | Medford, MA

August 2022 – May 2026

Captain, Long-Distance Runner

- Devote over 20 hours a week to practice, meetings with coaches, and competition throughout the year
- Meet with coaches and fellow captains to plan season objectives and discuss team direction, ethic and morale